

When Privatization Raises Prices: Refinery Divestment and Consumer Welfare in Brazil*

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Abstract

Privatization of state-owned enterprises is widely expected to lower prices and improve efficiency. We provide evidence that this presumption fails when privatized assets retain regional monopoly power. Exploiting the staggered divestment of two Brazilian oil refineries under a difference-in-differences design, we find that privatization raised refinery prices for all three main fuels. Net of taxes, and relative to mean 2025 prices, gasoline was overpriced by 7% (BRL 0.44 per liter; USD 0.08), liquefied petroleum gas (LPG) by 14% (BRL 15.56 per 13-kg cylinder; USD 2.78), and diesel by 4% (BRL 0.23 per liter; USD 0.04). For gasoline and LPG the overcharge passed through to consumers in full. For diesel, it was offset at retail by a concurrent tax reform that favored the treated states – absent which consumers would have faced higher diesel prices as well. Wholesale price data confirm that the overpricing originates at the refinery stage, not from retail margin expansion. We estimate a total consumer welfare loss of approximately BRL 10.1 billion (USD 1.8 billion) over nearly four years, with Amazonas bearing a per capita loss more than twice Bahia's. These losses are regressive, particularly for LPG: the poorest households spend more than ten times the income share of the wealthiest on this fuel.

Keywords: Privatization, refining market, welfare, difference-in-differences, fuel prices.

JEL Codes: R40, L12, L33.

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1 Introduction

In Brazil, the oil refining market has historically operated as a regional monopoly. Petrobras – a mixed-economy company jointly owned by the Brazilian government and private shareholders – held exclusive rights to oil exploration and refining until 1997. After liberalization, private entry remained negligible. Competing with an established state-owned incumbent was commercially unappealing, because Petrobras frequently set domestic prices below international benchmarks to serve political objectives, deterring new entrants. By 2020, Petrobras still controlled 98.6% of national refining capacity ([ANP, 2021](#)). Two reforms sought to change this. In 2016, Petrobras adopted an import parity policy (IPP), aligning domestic refinery gate prices with international benchmarks ([Petrobras, 2016, 2023](#)). In 2019, following an antitrust investigation into abuse of economic power, the company signed an agreement with Brazil’s competition authority (CADE) committing to divest half of its refining capacity – eight refineries in total ([CADE, 2019](#)).

This paper studies the welfare consequences of that divestment program. We focus on the first two refineries sold: one in Bahia, the country’s second-largest refinery with a 12% national market share, and one in Amazonas, the sole refinery in Brazil’s entire North region. These plants are canonical examples of local monopolies. The nearest competitor to the Bahia refinery is more than 800 kilometers away. For the Amazonas refinery, the figure exceeds 2,000 kilometers.¹ Both plants were privatized without price regulation, despite their structural features being more consistent with natural monopoly utilities than with competitive firms.

Our empirical strategy exploits the staggered timing of the two privatizations as a natural experiment. Using a difference-in-differences design ([Callaway & Sant’Anna, 2021](#)), we compare end-user fuel prices in Bahia and Amazonas (the treated states) against never-treated states outside the North and Northeast regions, before and after each divestment.² The privatization decisions were driven exclusively by government and antitrust considerations, with no influence from consumers or retail firms. Three fuels are examined: gasoline, liquefied petroleum gas (LPG), and diesel.

The results tell a simple story in three steps. First, privatization raised refinery prices for all three fuels. Our preferred estimates remove state fuel taxes from retail prices – reconstructing the tax schedule act by act – to isolate refinery pricing from fiscal policy. Net of taxes and in 2025 values, gasoline was overpriced by BRL 0.44 per liter (7% of the mean

¹See Figure 1. The two plants are Refinaria Landulpho Alves (RLAM) in Bahia and Refinaria Isaac Sabbá (REMAN) in Amazonas. Further details are provided in Section 2.

²We exclude the remaining North and Northeast states from the control group because, as neighbors of the treated states, they are potentially exposed to spillover effects of the privatizations.

price), LPG by BRL 15.56 per 13-kg cylinder (14%), and diesel by BRL 0.23 per liter (4%).³ Second, for gasoline and LPG the overcharge passed through to retail prices essentially in full and persistently, with no sign of mean reversion through June 2025, approximately 42 months after the first divestment. Third, where retail prices do not show the overcharge – diesel – taxation is the confounder, not the absence of overpricing. The treated states taxed diesel at the highest rate in the sample (18%), while most control states taxed it at the lowest (12%). The 2023 switch to a uniform national per-unit tax cut the former and raised the latter, masking the refinery overcharge in retail prices. The effects are uniformly larger in Amazonas, reflecting the Amazonas refinery’s stronger geographic isolation – five times Bahia’s overcharge for gasoline (BRL 1.18 vs. 0.23 per liter) and for diesel (BRL 0.65 vs. 0.13), and BRL 22.09 versus 12.22 per cylinder for LPG.

Wholesale prices paid by retailers rose alongside retail prices, confirming that the overpricing originates at the refinery stage rather than from retail margin expansion. The spatial pattern then reveals the market-power mechanism, which works differently across fuels according to how readily buyers can reach an alternative source. Gasoline moves cheaply in bulk by tanker or pipeline, so distributors arbitrage it within a state and the overcharge does not fall as stations approach a competing refinery: it is similar across Bahia – around R\$ 0.23–0.27 per liter, whether or not a rival is nearby – but five times larger in Amazonas, where the entire state lies beyond the reach of competing refineries. LPG moves in individual 13-kg cylinders – a unitized, labor-intensive process that makes interregional arbitrage costly – and its buyers are households with no practical alternative. For LPG, market power is correspondingly local: the overcharge is large where the privatized refinery is the only nearby supplier (its exclusive market) and statistically indistinguishable from zero where a rival refinery lies at a comparable distance (a shared market).⁴ Diesel travels in bulk as easily as gasoline, yet its discipline comes from the buyer rather than the product itself. Diesel is purchased mainly by commercial fleets that can source fuel from an alternative refinery wherever one is within range. This buyer mobility makes the diesel overcharge the smallest of the three and confines it to exclusive markets – modest in Bahia, where the Southeast’s refineries are reachable, and largest in Amazonas, where no alternative exists within thousands of kilometers.

The welfare implications are large. We estimate a total consumer welfare loss of approximately BRL 10.1 billion (USD 1.8 billion) between December 2021 and June 2025, BRL 5.7 billion in Bahia and BRL 4.4 billion in Amazonas. The aggregate figures, however,

³USD conversions use the 2025 average commercial exchange rate of BRL 5.59 per USD ([Banco Central do Brasil, 2025](#)).

⁴Brazil’s LPG distribution market is further segmented by concession areas assigned to major distributors (Liquigás, Ultragaz, Copagaz), limiting competitive pressure from outside the region.

understate how uneven the burden is. Amazonas holds roughly 2% of Brazil’s population, against about 7% for Bahia, yet its consumers absorbed a per capita loss of approximately R\$ 1,050 – more than two and a half times Bahia’s. These losses are regressive and land precisely in the North and Northeast, Brazil’s poorest regions. The distributional evidence is detailed in Section 2 and the welfare analysis in Section 4.

We add to the literature on the welfare effects of privatization. The dominant finding – that private ownership improves productive efficiency and need not harm consumers – rests on settings where competition or regulation disciplines pricing after divestment (Megginson & Netter, 2001; Shleifer, 1998). Our results speak directly to the argument from Vickers & Yarrow (1991) that post-privatization market structure matters more for consumers than the ownership change itself, and to Estache et al.’s (2001) observation that regulatory capacity is weaker in developing economies, making consumer-beneficial outcomes less likely. Previous evidence from Brazil’s electricity sector confirms the benign case, where privatization was consumer-neutral when the regulatory regime was adequate (Muller & Rego, 2021; Silvestre et al., 2010). We provide the complementary finding – a natural experiment in which a structural local monopoly was privatized without price regulation, producing the opposite of the intended outcome, as antitrust-mandated divestments designed to create competition instead transferred structural rents from consumers to private owners.

The paper advances the empirical literature on fuel and energy market power. We identify market power at the upstream refinery level and trace its transmission across a geographically dispersed retail network. A parallel wholesale price analysis rules out retail margin expansion as an alternative channel, and the spatial pattern provides a direct mechanism test, with overpricing large in each refinery’s exclusive market and statistically indistinguishable from zero where a rival refinery lies at a comparable distance. Relative to prior studies of Brazilian fuel demand, which estimate elasticities from state- or national-level panels (Santos, 2013; Cardoso et al., 2020; Cavalcanti & Jalles, 2013; Cardoso et al., 2019), we recover demand at the finer city-month level for the two treated states, the input to our welfare calculation.

Finally, the paper speaks to the literature on distributional consequences of reforms in energy markets in developing economies (Hipólito et al., 2025; Moz-Christofoletti & Pereda, 2021). The absence of price regulation at privatization embeds a permanent cost-of-living premium for the most exposed and least affluent consumers.

2 Background

The oil refining market in Brazil traces its roots to the 1930s, when small private refineries emerged alongside growing domestic demand. The sector was consolidated under state control in 1953 with the creation of Petrobras, which assumed a legal monopoly over oil exploration and refining. A key moment in this history was the inauguration of Refinaria Mataripe, later renamed the Landulpho Alves-Mataripe Refinery (RLAM), which became the main source of refining capacity in the Northeast.⁵ Until 1997, Petrobras held an uncontested legal monopoly. After liberalization, private entry remained negligible: competing with an established state-owned incumbent whose pricing policy frequently set domestic prices below international benchmarks was commercially unattractive. As of 2020, Petrobras still controlled 98.6% of national refining capacity (ANP, 2021). Bridgman et al. (2011) argues that this dominance did not necessarily imply inefficiency, as the credible threat of entry induced productivity improvements within the firm.

High market concentration prompted recurring calls for reform, documented across news coverage (Ordonez & Rosa, 2018), academic research (Rojas & Leite, 2018; Cabral, 2020), and government reports (CNPE, 2019; MME, 2017). A central challenge, however, was how to increase the number of market participants without replicating the monopoly structure at the regional level. As Tavares et al. (2006) notes, Brazil's refinery network was built to maximize production scale economies while minimizing distribution costs – a configuration that naturally produces geographic market segmentation. This concern is especially acute in the North and Northeast regions, as illustrated in Figure 1.

Two reforms were implemented to reduce Petrobras's dominance. First, in October 2016, Petrobras adopted the import parity policy (IPP), aligning refinery gate prices with international benchmarks (Petrobras, 2016, 2023). The IPP was discontinued in May 2023 following a change in federal administration. Two considerations drove its reversal: the mechanical link to international benchmarks exposed domestic consumers to sharp price volatility during oil shocks, and the incoming government sought to recover discretionary capacity to manage fuel prices – a tool that responds as much to electoral incentives as to stabilization objectives.⁶ (Petrobras, 2016, 2023).

The second reform was a 2019 agreement between Petrobras and CADE⁷ requiring

⁵For a comprehensive review of the historical development of Brazil's refining market, see Mendes et al. (2018) and CADE (2022b).

⁶The political appeal of fuel price management is not ideologically bounded in Brazil: price freezes (Rousseff), import parity (Temer, Bolsonaro), its reversal (Lula), and a federal subsidy of state fuel taxes (Bolsonaro) suggest that each incoming administration finds its own instrument – but the underlying electoral incentive remains constant.

⁷CADE – Conselho Administrativo de Defesa Econômica – is Brazil's antitrust authority.

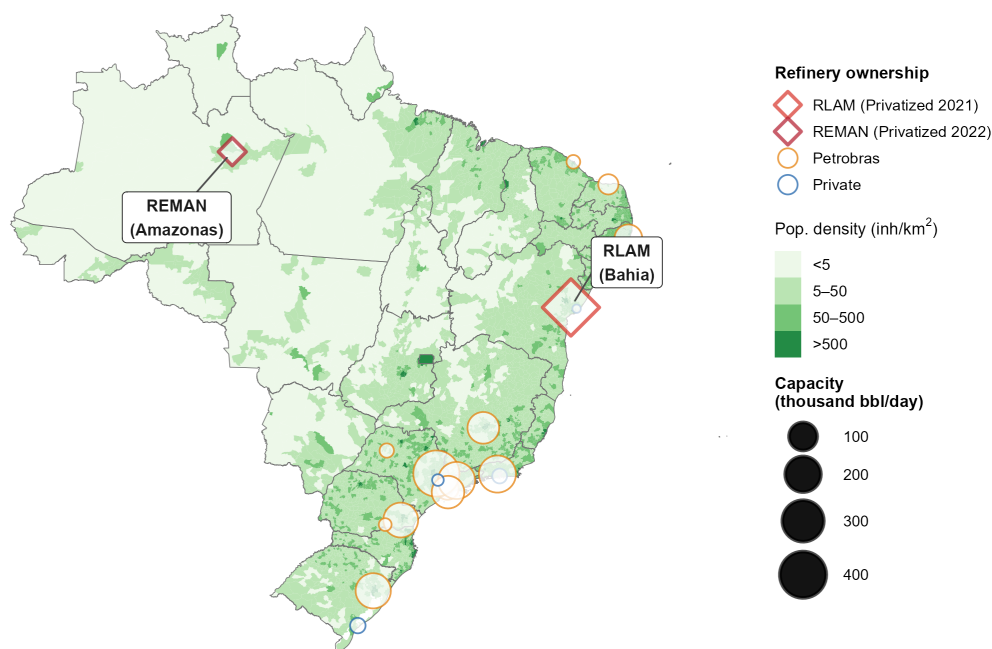


Figure 1: Refining Capacity and Population Density in Brazil (2022)

Notes: The two privatized refineries are shown as red diamonds – RLAM in Bahia (Mubadala Capital, 2021) and REMAN in Amazonas (Atem Group, 2022) – while all other refineries are shown as circles, orange for Petrobras-owned and blue for privately operated. Marker size is proportional to refining capacity (thousand barrels per day). Green shading represents population density by municipality in 2022 (people per km²). *Source:* ANP (2025) for refineries and IBGE (2023a) for population density.

Petrobras to divest half of its refining capacity, selling 8 of its 13 refineries. The agreement followed a 2017 antitrust investigation into market power abuse. In July 2021, CADE approved the first transaction, the acquisition of the Bahia refinery by the Emirati group Mubadala Capital, with operational control transferring in December 2021 (CADE, 2022a). The Amazonas refinery's sale contract was signed in August 2021, and the Atem Group assumed operational control in December 2022. These were the only two transactions completed. In May 2024, Petrobras and CADE cancelled the remaining six divestments, as evidence mounted that the program was not delivering the competitive benefits originally intended.⁸

The two plants differ markedly in market power and geographic reach. The Bahia refinery commands a 12% share of national refining capacity and is the dominant supplier in its state. Its nearest competing refinery is more than 805 kilometers away, and the combined capacity of the other two Northeast refineries (Lubnor and RNEST) amounts to

⁸Petrobras and CADE signed an addendum to the 2019 divestment agreement releasing Petrobras from the obligation to sell the remaining six refineries (CADE & Petrobras, 2024).

only one-third of it ([ANP, 2022](#)). The Amazonas refinery is the sole refinery in Brazil's entire North region, with no competitor within 2,000 kilometers, serving a geographically isolated area connected almost exclusively by river transport. Despite its modest 1.7% national market share, it exercises strong regional monopoly power, a structural feature that foreshadows the larger overpricing we document in Amazonas relative to Bahia.

The Amazonas refinery's monopoly is not absolute, because distributors in the North can also source fuel through imports and cabotage shipments arriving at river terminals in the Manaus area ([ANP, 2022](#)). In practice, however, the long river logistics chain and port costs make imported fuel an expensive alternative, so import parity operates as a high price ceiling rather than an effective competitive constraint. This distinction matters for interpreting our results, because a profit-maximizing private owner can raise prices toward the import-parity ceiling – pricing room that the state-owned incumbent, bound by nationwide pricing policies, had not exploited.

The North and Northeast are also Brazil's poorest regions. In 2022, per capita income in both regions stood at roughly two thirds of the national average ([IBGE, 2023b](#)), so the price impacts of these divestments fall disproportionately on lower-income consumers.

The distributional stakes are sharpened further by how different fuels figure in household budgets, a margin that distributional studies of energy pricing in Brazil have repeatedly stressed ([Hipólito et al., 2025](#); [Garaffa et al., 2021](#); [Moz-Christofolletti & Pereda, 2021](#)). LPG – sold in 13-kg cylinders and used primarily for cooking – is a necessity with no affordable substitute for most Brazilian households, particularly those without access to piped natural gas. Data from the 2017–2018 Household Budget Survey (POF) show that the poorest households allocate 2.19% of total expenditure to LPG, compared to just 0.21% for the wealthiest – a ratio exceeding 10 to one. This asymmetry reflects a familiar pattern in developing economies, where rising income leads households to substitute away from bottled gas toward piped natural gas or restaurant meals, compressing LPG's budget share among the affluent. Gasoline exhibits a similar but less pronounced gradient – 2.11% for the poorest versus 0.37% for the wealthiest, a ratio of 5.7 to one – consistent with higher-income households owning more vehicles but spending a smaller fraction of their budgets on fuel. Diesel is excluded from this household analysis because Brazilian regulation prohibits its use in light vehicles, so household exposure to diesel prices is largely indirect, embedded in freight and transportation costs. Figure 9 in the Appendix displays the full budget-share gradient across the seven income classes.

Opponents of privatization would predict higher end-user prices, while proponents would expect lower prices. Figure 11 offers preliminary evidence consistent with the pessimistic view. In Bahia, LPG prices shifted from 8% below the control group before

privatization to 4% above it thereafter. In Amazonas, prices were already elevated at 7% above the control before the sale and surged to 18% above following privatization. Analogous patterns for gasoline and diesel are shown in Figures 7 and 8 in the Appendix.

While these visual patterns are suggestive, informal inspection cannot rule out confounding factors. Of particular concern is the sequence of reforms to the ICMS – the state-level value-added tax on fuel – which began in June 2022, roughly four months before Brazil’s presidential election, when a federal law capped the state-varying ad valorem rates, and culminated in mid-2023, when those rates were replaced by a single nationwide per-unit tax for each fuel. The next section employs difference-in-differences methods to isolate the causal effect of privatization from such confounders.

3 Price Effects of Privatization

We begin by outlining our empirical strategy, then present baseline results estimating the average treatment effect of refinery privatizations, followed by analyses of cohort heterogeneity and of spatial heterogeneity across exclusive and shared markets.

3.1 Empirical Strategy and Data

To measure the price effects of privatization, we employ a difference-in-differences (DiD) research design using the estimator proposed by [Callaway & Sant’Anna \(2021\)](#). This estimator is particularly suited to our setting, which involves staggered treatment adoption (the Bahia refinery privatized in December 2021 and the Amazonas refinery in December 2022), as illustrated in Figure 5 in the Appendix. The [Callaway & Sant’Anna](#) estimator addresses potential biases associated with staggered adoption by computing group-time average treatment effects and then aggregating them appropriately ([De Chaisemartin & D’haultfœuille, 2023](#); [Roth et al., 2023](#); [Baker et al., 2022](#)).

The fundamental assumption underlying our identification strategy is conditional parallel trends: in the absence of privatization, fuel prices in Bahia and Amazonas would have followed the same trend as prices in the control group (never-treated states, excluding the North and Northeast). We test this assumption using the framework of [Egami & Yamauchi \(2023\)](#), which exploits multiple pre-treatment periods, applying the test to the net-of-tax outcome that is the focus of our preferred specification. The results, presented in Figures 2, 3, and 4 in the Appendix, reveal no observable pre-trends in the gasoline, LPG, or diesel markets.

Formally, for each group g (cohort defined by treatment timing) and time t , we estimate:

$$ATT(g, t) = \mathbb{E} [P_{it}(g) - P_{it}(0) \mid G_i = g] \quad (1)$$

where P_{it} is the real fuel price at unit i in period t , $G_i = g$ indicates that unit i was first treated in period g , and $P_{it}(0)$ is the counterfactual price absent privatization. We use fuel retailers in never-treated states as the comparison group. We report three aggregations of the $ATT(g, t)$ following [Callaway & Sant'Anna \(2021\)](#): the overall effect (a single average weighted by each cohort's share of post-treatment observations), the group-specific effects (separate averages for the Bahia and Amazonas cohorts, reported in Section 3.2), and the dynamic event-study effects (averages by length of exposure, shown in the event-study figures).

Our net-of-tax strategy depends on how fuel taxation is structured in Brazil, so a brief overview is warranted. Taxes are levied at three levels of government, but only one of them threatens our identification. There is no municipal fuel tax, so within-state price variation across cities is tax-free. Federal taxes (PIS/Cofins and CIDE) are uniform across all fuel retailers nationwide, so they are fully absorbed by the week fixed effects and require no adjustment. The state-level tax (the ICMS), by contrast, varies across states and over time, and – crucially for our design – can distort retail prices in ways that week fixed effects do not capture.

We estimate the model on three outcomes. The first is the retail price – what households actually paid. The second, our preferred outcome, is the *net-of-tax* price – the retail price net of the state fuel tax (ICMS) – which measures what the refinery charges and therefore speaks directly to market power. The third is the wholesale price paid by fuel retailers to acquire fuel from refineries or distributors, which locates where in the supply chain the overpricing originates. The distinction between the first two matters because the ICMS changed substantially and asymmetrically across states during the post-treatment period.

Two reforms are relevant. First, on June 23, 2022, Complementary Law 194 capped the state fuel tax at each state's standard rate of 17–18%. This was binding mainly for gasoline, which had been taxed at 25–34%. Second, the ad valorem state rates were replaced by a single nationwide per-unit tax – in May 2023 for diesel and LPG, and in June 2023 for gasoline. Because the treated states taxed diesel at the top rates in the sample while most control states taxed it at the bottom, these reforms mechanically compressed retail-price gaps between treated and control states, masking the refinery overcharge in retail prices.

To construct the net-of-tax outcome, we reconstruct the state-by-week tax schedule act by act – every state rate, the federal cap, and the per-unit rates with their adjustments – as

documented in Tables 7 and 8 in the Appendix. In the ad valorem regimes, the ICMS is levied on the tax-inclusive price, so the net price is $P(1 - \tau_{st})$, while in the per-unit regime it is the retail price minus the ad rem rate.⁹

The end-user data come from the *Sistema de Levantamento de Preços* of the ANP (ANP, 2025), released weekly from January 2019 to June 2025. Each observation records the retailer name, CNPJ, full address, and end-user prices for gasoline, diesel, LPG, and ethanol. In our preferred specification, we exclude states from the North and Northeast regions to account for potential spillover effects of the privatization on neighboring markets. To address pre-treatment heterogeneity in the panel, we restrict the sample to retailers observed for at least six months before the earliest treatment date. Figure 6 in the Appendix maps the resulting estimation sample, 31,637 retailers across 366 municipalities. The ANP survey follows a structured sampling plan rather than a census of Brazilian fuel retailers, so our sample does not cover every retailer in the country. It nonetheless provides broad geographic coverage of the treated states and of the control group.

The wholesale price data, used to locate where in the supply chain the overpricing arises, come from two sources. For gasoline and diesel, we use confidential invoice-level (NF-e) records shared by the ANP under a data agreement, covering January 2019 to December 2023. For LPG, we use the publicly available distributor prices from the ANP's *Preços de Distribuição de Combustíveis* (PDC) program, reported at the municipality-week level for 459 municipalities between August 2020 and April 2026. These series let us test whether the overpricing originates upstream, at the refinery and distribution stage, rather than in retailer margins.

Table 1 reports summary statistics for the estimation samples: end-user and wholesale prices for gasoline, diesel, and LPG.

⁹The wholesale analysis in Panel B of Table 2 provides a further check independent of retail taxation.

Table 1: Summary Statistics – Estimation Samples

	N	Mean	Std. Dev.	Min	Max
<i>Panel A – Gasoline (BRL/liter)</i>					
Price (retail, nominal)	962,009	5.473	0.983	2.918	8.990
Price (retail, real 2025)	962,009	6.386	0.863	3.944	10.250
Price (net of tax, real 2025)	962,009	4.809	0.611	2.984	7.993
Wholesale price (nominal) [†]	9,239,438	4.970	0.772	2.635	7.087
Treated	962,009	0.050	0.218	0	1
<i>Panel B – Diesel (BRL/liter)</i>					
Price (retail, nominal)	487,922	5.063	1.319	2.453	8.999
Price (retail, real 2025)	487,922	5.878	1.074	3.331	10.189
Price (net of tax, real 2025)	487,922	4.997	0.917	2.873	8.757
Wholesale price (nominal) [†]	4,244,327	4.757	1.065	2.160	7.156
Treated	487,922	0.055	0.228	0	1
<i>Panel C – LPG (BRL/13-kg cylinder)</i>					
Price (retail, nominal)	854,063	92.084	19.840	47.990	160.000
Price (retail, real 2025)	854,063	108.345	15.853	67.119	183.440
Price (net of tax, real 2025)	854,063	92.161	14.289	51.335	161.427
Wholesale price (nominal) [‡]	106,002	73.232	9.505	28.126	134.780
Treated	854,063	0.046	0.210	0	1

Notes: End-user rows (retail prices and net-of-tax) cover retailer-week observations from January 2019 to June 2025, with North and Northeast states excluded from the control group. Real prices are deflated by the IPCA to 2025 BRL. Treated equals 1 for a retailer-week observation in Bahia after December 2021 or in Amazonas after December 2022, so its mean is the share of treated post-period observations in the panel, not the share of treated retailers. [†] Wholesale gasoline and diesel prices are from a confidential ANP data-sharing agreement (NF-e invoice records, January 2019–December 2023). [‡] Wholesale LPG prices are municipality-week distributor prices from the public ANP PDC program (August 2020–April 2026; 30 municipalities in Bahia, 5 in Amazonas), converted from BRL/kg to BRL per 13-kg cylinder. Source: ANP (2025).

3.2 Results on the Price Effects of Privatization

Table 2 presents the price effects of privatization across the supply chain. Panel A reports the preferred net-of-tax estimates, which isolate the refinery overcharge from the asymmetric tax reforms. Panel B reports wholesale prices paid by fuel retailers, and Panels C and D report consumer (retail) prices. The headline result is in Panel A, where privatization

raised the refinery prices for *all three fuels*, with gasoline rising by R\$ 0.44 per liter, LPG by R\$ 15.56 per cylinder, and diesel by R\$ 0.23 per liter.

Table 2: Impact of Refinery Privatization on Fuel Prices

	Gasoline	LPG	Diesel
<i>Panel A – Net-of-Tax Prices (preferred)</i>			
ATT	0.441*** (0.016)	15.558*** (0.531)	0.234*** (0.018)
<i>Panel B – Wholesale Prices</i>			
ATT	0.154*** (0.019)	12.448*** (0.485)	0.023*** (0.005)
<i>Panel C – Retail Prices (excl. North & Northeast)</i>			
ATT	0.430*** (0.020)	15.162*** (0.607)	0.019 (0.019)
<i>Panel D – Retail Prices (all 27 states)</i>			
ATT	0.391*** (0.020)	15.013*** (0.544)	0.027 (0.020)
FE: retailer / city	✓	✓	✓
FE: week / time	✓	✓	✓

Notes: ATT from [Callaway & Sant’Anna \(2021\)](#) in real 2025 BRL, except Panel B gasoline and diesel (see below). The control group is never-treated states; Panels A, C, and B exclude the North and Northeast regions, while Panel D retains all 27 states. Net-of-tax prices remove the state ICMS from retail prices using the act-by-act schedule of Tables 7–8 in the Appendix. Panels A, C, and D use retailer and week fixed effects with multiplier-bootstrap standard errors clustered at the retailer level. In Panel B, gasoline and diesel use confidential wholesale prices from an ANP data-sharing agreement (2021–2023), estimated via the TWM estimator ([Wooldridge, 2021](#)) with city and time fixed effects and city-clustered standard errors (not updated to 2025 values in this revision); LPG uses public ANP distributor prices (PDC program, 459 municipalities, Aug. 2020–Apr. 2026), estimated via [Callaway & Sant’Anna \(2021\)](#). LPG unit is BRL per 13-kg cylinder, and all other prices are in BRL per liter. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

The net-of-tax gasoline overpricing of R\$ 0.44 per liter corresponds to 7% of the average 2025 gasoline price (R\$ 6.28), and the LPG overpricing of R\$ 15.56 per cylinder to 14% of the average 2025 LPG price (R\$ 108.22). For diesel, the net-of-tax overcharge of R\$ 0.23 per liter (about 4% of the average 2025 diesel price) is cleanly positive, and how much of it reached consumers at retail is a separate question, addressed below.

A natural concern is that retailers exploited the market turmoil of the privatization transition to widen their margins, so that the overpricing in Panel A would reflect retailer conduct rather than refinery pricing. Panel B addresses this by examining wholesale prices paid by fuel retailers – one step back in the supply chain. Wholesale gasoline prices also rose significantly (R\$ 0.15 per liter). For LPG, the wholesale distributor price increased by R\$ 12.45 per cylinder, about 80% of the end-user overpricing, confirming that the distortion originates at the refinery stage rather than from retail margin expansion.

Panels C and D close the picture with consumer prices. For gasoline and LPG, the overcharge reached households essentially in full, with retail prices rising by R\$ 0.43 per liter and R\$ 15.16 per cylinder (Panel C), nearly identical to the net-of-tax estimates in Panel A. For diesel, the retail-price ATT is statistically indistinguishable from zero (0.019), in contrast to the net-of-tax overcharge of R\$ 0.23 per liter, because the treated states taxed diesel at 18% (the highest rate in the sample) while most control states taxed it at 12%, so the 2023 single-stage reform cut the tax burden on treated-state diesel while raising it in the control states, masking the refinery overcharge in retail prices. The all-states specification in Panel D yields a slightly smaller gasoline coefficient (R\$ 0.39 vs. R\$ 0.43), a pattern consistent with pricing spillovers to neighboring North and Northeast states, and thus supportive of excluding those states in the preferred sample.

Heterogeneous Effects Across Cohorts

Two key questions arise after establishing the average effects: (i) Did consumers in Bahia experience different overpricing compared to those in Amazonas? (ii) Are the effects persistent or merely short-term?

Table 3 and Figures 12 and 13 address these questions.

The net-of-tax estimates reveal a strikingly consistent pattern, with the overpricing positive for every fuel in both states and always larger in Amazonas. For gasoline, suppliers in Amazonas overcharge R\$ 1.18 per liter, five times the R\$ 0.23 in Bahia; for diesel, R\$ 0.65 against R\$ 0.13, again a factor of five; for LPG, R\$ 22.09 against R\$ 12.22 per cylinder. This ranking is exactly what the Amazonas refinery's stronger monopoly position predicts, as it is the sole refinery in the North region, geographically isolated by over 2,000 kilometers from any competitor, whereas the Bahia refinery faces some degree of competition from refineries in the Southeast through interstate fuel transport.

The consumer-price rows show how much of each overcharge reached households. For gasoline and LPG, pass-through was essentially complete in both states. For diesel, the two outcomes diverge, with Amazonas consumers absorbing most of the overcharge

(R\$ 0.34 retail vs. R\$ 0.65 net of tax), while in Bahia retail diesel prices actually edged down (−R\$ 0.06) despite the positive net-of-tax effect. The wedge is the 2023 tax reform, because both treated states taxed diesel at 18% (the top of the sample), so the switch to the uniform national per-unit rate cut their diesel tax burden while raising taxes in the 12% control states. In Bahia the tax relief happened to offset the refinery overcharge at retail, while in Amazonas the overcharge was too large to be masked.

Diesel also illustrates how buyer mobility shapes the overcharge. Its buyers are predominantly commercial fleets, which respond to a price gap not by consuming less but by sourcing fuel from whichever refinery is within reach. In Bahia they can refuel across state borders, so the privatized refinery faces a more elastic residual demand and posts the smallest overcharge in the table. In Amazonas, where fuel moves almost exclusively by river, even these fleets have no alternative refinery nearby, and the overcharge is correspondingly large. The same logic operates within Bahia, where it sorts retailers into the exclusive and shared markets analyzed in Section 3.3.

Table 3: Impact of the Bahia and Amazonas Refinery Privatizations by Cohort

	<i>Bahia</i>			<i>Amazonas</i>		
	Gasoline	LPG	Diesel	Gasoline	LPG	Diesel
<i>Net-of-tax prices (preferred)</i>						
ATT	0.234*** (0.013)	12.221*** (0.706)	0.129*** (0.019)	1.177*** (0.039)	22.093*** (0.675)	0.647*** (0.037)
<i>Consumer (retail) prices</i>						
ATT	0.200*** (0.017)	14.302*** (0.829)	−0.063*** (0.021)	1.248*** (0.047)	16.845*** (0.753)	0.341*** (0.037)
FE: retailer	✓	✓	✓	✓	✓	✓
FE: week	✓	✓	✓	✓	✓	✓

Notes: Group-specific ATTs from [Callaway & Sant’Anna \(2021\)](#). Bootstrap standard errors in parentheses. LPG unit is BRL per 13-kg cylinder. Net-of-tax prices strip the state ICMS from retail prices using the act-by-act schedule of Tables 7–8 in the Appendix. The control group is never-treated states, excluding the North and Northeast. Net of taxes, the overpricing is positive for every fuel in both cohorts and uniformly larger in Amazonas. In retail prices, the 2023 fuel-tax reform – which cut the treated states’ 18% diesel rate to the uniform national per-unit rate while raising rates in the 12% control states – offsets the diesel overcharge in Bahia (−0.063) and partially masks it in Amazonas. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

A key concern in both markets is whether overpricing is fading or persisting. The dynamic event study plots show no clear trend toward zero through June 2025 – approximately 42 months after the Bahia refinery’s privatization and 30 months after the Amazonas refinery’s. Note that the event studies aggregate both treatment cohorts in event time. Because Amazonas was privatized one year after Bahia, the longest post-treatment horizons (beyond roughly 130 weeks, i.e., around period 150 in Figure 12) are identified by the Bahia cohort alone. The apparent decline in the gasoline effect at the end of the event-time window is therefore compositional – the Amazonas cohort, which faces the largest overpricing, exits the estimation sample at those horizons – rather than evidence of mean reversion. This absence of mean-reversion suggests that the overpricing is a lasting structural feature of the privatized local monopoly arrangement, rather than a short-run adjustment.

3.3 Spatial Heterogeneity: Exclusive versus Shared Markets

The overpricing should not be uniform within a state. A refinery can charge above the competitive price only where its buyers have no comparable alternative: if a rival refinery sits at a similar distance, local distributors can source the same fuel from either plant, and that outside option disciplines the price regardless of who owns the privatized refinery. Privatization should therefore raise prices where the divested refinery is the only nearby supplier, and leave them unchanged where a competitor is equally close. This logic delivers a test of the market-power mechanism that stays within treated states. We split treated retailers into those for which the privatized refinery is effectively the sole nearby supplier and those that a rival refinery could plausibly serve, and ask whether the overpricing is concentrated in the former.

Keeping the comparison internal to each state has two virtues. It requires no never-treated control group, since both groups underwent the same privatization. And it lets diesel be read on the same footing as the other fuels, because every estimate in this subsection uses net-of-tax prices, following our preferred specification (Table 2, Panel A), which strips out the staggered ICMS reforms that confounded the aggregate diesel result in Section 3.1. The within-state contrast then isolates local market power from any statewide cost or demand shock.

For each retailer we compute how much closer the privatized refinery is than the next-nearest one – its monopoly margin – measured among the ten refineries operating in Brazil. When this margin is large, the nearest competitor is far away and the privatized refinery is effectively the only low-cost supplier. When it is near zero, two refineries are roughly equidistant and either could serve the retailer. We label a treated retailer an exclusive

market when the privatized refinery is its nearest by a margin of at least 100 km, and a shared market when that margin falls below 100 km.

This classification first validates the state-level treatment definition. The privatized refinery is the nearest supplier for 97.5% of Bahia's retailers and for all of Amazonas's, and no treated retailer is closer to a competing refinery, so an economically defined placebo group does not exist. In Amazonas the isolation is extreme, with the second-nearest refinery between 1,652 and 2,385 km farther away than the Amazonas refinery, so every Amazonas retailer is an exclusive market. Shared markets are therefore confined to southern Bahia, where the Bahia refinery and Regap (Minas Gerais) lie at comparable distances.

Table 4 reveals a sharp gradient for the two fuels that travel poorly. For LPG the contrast is striking, with retailers in the Bahia refinery's exclusive market bearing the full overpricing (R\$ 12.9 per cylinder), whereas retailers in shared markets show an effect statistically indistinguishable from zero (R\$ 0.93, SE 2.51) and significantly smaller than the exclusive-market effect. Where a rival refinery is available at a comparable distance, the privatized refinery loses its ability to overprice bottled gas, the defining signature of local market power.

The same test resolves the diesel puzzle from Section 3.1. There the aggregate diesel effect was muddled by the staggered ICMS reforms. Here, with net-of-tax prices and a comparison internal to each state, the market-power reading is clean. In the exclusive Amazonas market, where the Amazonas refinery's nearest competitor lies more than 1,600 km away, diesel rose by R\$ 0.65 per liter, the same exclusive-market overpricing we find for gasoline and LPG. Within Bahia the exclusive-market effect is positive and significant (R\$ 0.13), while the small shared-market group shows a smaller effect that is not statistically distinguishable from zero (R\$ 0.06), and the within-state gradient points the same way as LPG, though the handful of shared-market diesel retailers (41) makes it less precisely estimated. Diesel, in short, was never an exception. Once taxes are netted out, its overpricing is concentrated – exactly as the mechanism predicts – where the refinery faces no nearby rival.

Gasoline behaves differently. Shared markets show no attenuation (R\$ 0.27 vs. R\$ 0.23 in Bahia), which fits the fact that gasoline is far more transportable than bottled LPG, so arbitrage smooths within-state price differences and gasoline market power operates across states rather than within them, with the effect in Amazonas more than five times Bahia's.

Table 4: ATT by Market Classification: Exclusive vs. Shared Markets (Bahia)

	<i>Gasoline</i> (R\$/liter)	<i>Diesel</i> (R\$/liter)	<i>LPG</i> (R\$/cylinder)
<i>Bahia</i>			
Exclusive market (margin ≥ 100 km)	0.226*** (0.013)	0.130*** (0.019)	12.863*** (0.546)
Shared market (margin < 100 km)	0.272*** (0.021)	0.055 (0.041)	0.930 (2.510)
Frequency	weekly	weekly	monthly

Notes: Each cell reports the ATT from a separate [Callaway & Sant’Anna \(2021\)](#) estimation on net-of-tax real 2025 prices (tax schedule as in Table 2, Panel A); the control group is never-treated states, excluding the North and Northeast. The table is restricted to Bahia, the only treated state with a within-state contrast. Amazonas is omitted because every Amazonas retailer is an exclusive market – the nearest rival refinery is more than 1,600 km away – so it offers no shared-market comparison; estimated on its exclusive retailers alone, the Amazonas overcharge is R\$ 1.177 per liter for gasoline, R\$ 0.647 for diesel and R\$ 21.860 per cylinder for LPG, the same pattern found in Bahia’s exclusive market. The monopoly margin is the difference between a retailer’s distance to the second-nearest and to the nearest of the ten operational Brazilian refineries (Haversine). Retailers in Bahia: gasoline – 920 exclusive and 53 shared; diesel – 664 exclusive and 41 shared; LPG – 1,099 exclusive and 83 shared. Gasoline and diesel are estimated weekly and LPG at monthly frequency (retailer-month median), because weekly ANP price collection is too sparse in shared-market municipalities. The shared-market LPG effect is significantly smaller than the exclusive-market effect ($p < 0.01$). The 100-km cutoff is not knife-edge, since no treated retailer has a margin between 88 and 150 km, so any cutoff in that range yields an identical classification; with a stricter 50-km cutoff the shared-market estimates are R\$ 0.219 (SE 0.023) for gasoline, R\$ 0.128 (SE 0.043) for diesel and R\$ –2.102 (SE 2.458) for LPG. Bootstrap standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$; * $p < 0.10$.

4 Welfare Analysis

The analysis thus far has demonstrated persistent overpricing in end-user fuels caused by the Bahia and Amazonas refinery privatizations. The next step is to gauge the welfare effects of this policy.

4.1 Fuel Demand in Bahia and Amazonas

Numerous studies have estimated fuel demand in Brazil, typically using state- or national-level panels ([De Freitas & Kaneko, 2011](#); [Santos, 2013](#); [Junior, 2013](#); [Cardoso et al., 2019](#),

2020). Our contribution is to estimate demand at the finer city (municipality) level, using a monthly city-level panel for Bahia and Amazonas, the two states directly affected by the privatizations. This finer resolution matters for the welfare calculation, because the relevant demand response is the one in the treated markets themselves, not a state or national average that blends them with very different markets.

We use Two-Stage Least Squares (2SLS) to address the well-known simultaneity between price and quantity demanded. The first and second stages are:

$$\text{First Stage: } P_{it}^f = \theta + \phi Z_{it}^f + \eta \text{Inc}_{it} + \kappa_t + \omega_i + v_{it} \quad (2)$$

$$\text{Second Stage: } Q_{it}^f = \alpha + \delta \hat{P}_{it}^f + \beta P_{it}^a + \gamma \text{Inc}_{it} + \lambda_t + \mu_i + \varepsilon_{it} \quad (3)$$

where Q_{it}^f is the log quantity (cubic meters) of fuel f sold in municipality i at time t ; P_{it}^f is the log fuel price; P_{it}^a is the log price of the substitute fuel (gasoline for ethanol and vice versa; no substitute for diesel); Inc_{it} is municipal income; and κ_t, ω_i are time and municipality fixed effects. The instrumental variables Z_{it}^f follow the Hausman instrument approach (Berry & Haile, 2021), using contemporaneous prices of the same fuel in other geographic regions, international oil prices, and sugar prices, each interacted with the distance to the closest refinery.

Table 5 presents the demand estimates for Bahia and Amazonas. All three fuels are price-inelastic, as expected for goods with few short-run substitutes, but they differ in degree. Gasoline is the most price-responsive (own-price elasticity of about -0.72), followed by LPG (-0.66) and diesel (-0.54); the comparatively rigid diesel demand fits its use by commercial fleets whose activity levels are hard to adjust in the short run. The income responses are more revealing. Gasoline and diesel demand rise with municipal income (elasticities of 0.52 and 0.48), whereas LPG demand is essentially flat in income (a coefficient of 0.01, statistically indistinguishable from zero). The interpretation is clear, because cooking gas is a necessity consumed in a roughly fixed quantity regardless of income, so richer households do not buy more of it – unlike motor fuels, whose consumption scales with vehicle ownership and travel as income grows. This is also why the LPG overcharge is so regressive, falling on a quantity that poor and rich households consume alike. The own-price elasticities are used to compute the indirect welfare effect (area B in Figure 14).

4.2 Welfare Effects in Bahia and Amazonas

We assess changes in consumer surplus following Hausman & Newey (1995); Dargay & Goodwin (1995). As illustrated in Figure 14, the welfare loss from a supply-side price shift

Table 5: Fuel Demand Estimates in Bahia and Amazonas (City-Month, 2SLS)

	Gasoline	Diesel	LPG
Gasoline price (log)	−0.719** (0.282)		
Diesel price (log)		−0.535*** (0.074)	
LPG price (log)			−0.660*** (0.180)
Income proxy (log)	0.515*** (0.007)	0.475*** (0.010)	0.010 (0.005)
Num. Obs.	3,588	3,835	3,196
R^2	0.590	0.376	0.180
Instruments			
Prices (neighbors, log)	✓	✓	✓
Distance to refinery	✓	✓	
Global oil price	✓	✓	
Global sugar price	✓		

Notes: Log-log specification; parameters are elasticities. Standard errors in parentheses. ** $p < 0.05$; *** $p < 0.01$. LPG elasticity estimated via 2SLS on a municipality-month panel for Bahia and Amazonas covering 2013–2020 (3,196 obs), extending the sample beyond the 2019–2020 window available in the main price dataset. The instrument follows the same Hausman approach used for gasoline and diesel (prices in other regions).

is:

$$\Delta CS = \underbrace{\Delta P \cdot Q_2}_{\text{Area A}} + \underbrace{\frac{\Delta P \cdot \Delta Q}{2}}_{\text{Area B}} \quad (4)$$

where ΔP is the privatization-induced effect on consumer prices – the net-of-tax cohort ATT from Table 3 passed through the tax regime in force each month, since under the ad valorem ICMS a producer overcharge of Δ raises the retail price by $\Delta/(1 - \tau_{st})$ and under the 2023 single-stage regime it passes one-for-one. Q_2 is the quantity sold after privatization, and ΔQ is the demand response estimated from Table 5.

Table 6 presents the welfare calculations for the three fuels across both privatizations, using the preferred net-of-tax cohort effects from Table 3. The consumer cost of privatization in each month is the net-of-tax overcharge passed through the tax regime actually in force: under the ad valorem regimes the ICMS is levied on the tax-inclusive price, so a producer overcharge of Δ raises the retail price by $\Delta/(1 - \tau_{st})$, while under the single-stage regime the pass-through is one-for-one. We therefore multiply each cohort ATT by a volume-weighted regime factor (between 1.03 and 1.11 across fuel-cohort cells) before applying it

to the volumes sold. The results indicate a total consumer loss of approximately BRL 10.1 billion between December 2021 and June 2025 (Area A + Area B). Area A alone (direct effect) amounts to BRL 9.8 billion. The indirect demand-reduction effects (Area B) add BRL 158 million for gasoline ($|\varepsilon| = 0.719$), BRL 94 million for LPG (using our estimated price elasticity of $|\varepsilon| = 0.660$), and BRL 57 million for diesel ($|\varepsilon| = 0.535$). Of the BRL 10.1 billion, about BRL 9.2 billion is the transfer to the supply chain itself; roughly BRL 0.6 billion is the additional amount consumers paid through the ad valorem amplification of the overcharge, collected as state tax revenue; and BRL 0.3 billion is deadweight loss. These estimates exclude spillovers to other products (naphtha, kerosene) and to neighboring states in the North and Northeast.

Table 6: Welfare Variation by Fuel Following Privatization Sales (BRL millions, 2025 values)

	<i>Bahia</i>			<i>Amazonas</i>			Total
	Gasoline	LPG	Diesel	Gasoline	LPG	Diesel	
Area A (direct)	2,160	1,748	1,735	1,968	466	1,726	9,803
Area B – Gasoline ($ \varepsilon = 0.719$)	32	–	–	126	–	–	158
Area B – LPG ($ \varepsilon = 0.660$)	–	65	–	–	29	–	94
Area B – Diesel ($ \varepsilon = 0.535$)	–	–	10	–	–	47	57
Total	2,192	1,813	1,745	2,094	495	1,773	10,112

Notes: Area A is $\Delta P \times Q_2$, where ΔP is the net-of-tax cohort ATT of Table 3 multiplied by the volume-weighted regime factor (the factor equals $1/(1 - \tau_{st})$ in the months under the ad valorem regime and 1 under the single-stage regime; resulting consumer-price effects: gasoline 0.260 in BA, 1.226 in AM; LPG 12.90 and 22.84 per cylinder; diesel 0.139 and 0.669). Volumes are total sales over the post-privatization period (Dec 2021–Jun 2025 for Bahia and Dec 2022–Jun 2025 for Amazonas), measured from the transfer of operational control: gasoline 8.31 and 1.60 billion liters and diesel 12.50 and 2.58 billion liters from ANP open sales data (BA and AM, respectively); LPG 135.4 and 20.4 million 13-kg cylinders. Area B – Gasoline uses $|\varepsilon| = 0.719$ estimated in Table 5. Area B – LPG uses $|\varepsilon| = 0.660$, estimated via 2SLS on a municipality-month panel for Bahia and Amazonas (2013–2020, 3,196 obs); see Table 5. Area B – Diesel uses $|\varepsilon| = 0.535$ from Table 5. All values in millions of real 2025 BRL; totals may differ from the sum of rounded components. Script: `run_welfare_net.R`.

The consumer loss is approximately BRL 5.7 billion in Bahia and BRL 4.4 billion in Amazonas. The comparison understates how uneven the burden is, because Bahia has roughly 14 million inhabitants and Amazonas only 4 million, so the per capita loss in Amazonas – about R\$ 1,050 – is more than two and a half times Bahia’s. This sharpens the central message of the paper. Amazonas, a small, remote market representing only about 2% of Brazil’s population but where the privatized refinery faces no competitive discipline in any fuel, bears by far the heaviest per capita welfare loss.

5 Conclusion and Policy Implications

Contrary to the belief that privatization always leads to lower prices, our findings show the opposite in this setting. Privatization raised the refinery prices for all three fuels. Net of taxes, gasoline was overpriced by R\$ 0.44 per liter (7% of the 2025 mean price), LPG by R\$ 15.56 per 13-kg cylinder (14%), and diesel by R\$ 0.23 per liter (4%). For gasoline and LPG the overcharge passed through to consumers essentially in full. For diesel, the retail-price effect is masked by the 2023 fuel-tax reform, which cut the treated states' top-of-sample diesel rates while raising rates in the low-tax control states. The overpricing is uniformly larger in Amazonas than in Bahia – five times larger for gasoline and diesel – reflecting the Amazonas refinery's stronger geographic isolation. The spatial analysis pins down the mechanism, as within Bahia LPG overpricing is confined to the privatized refinery's exclusive market, while retailers in shared markets – where a rival refinery lies at a comparable distance – show no significant effect. It is local monopoly power, not private ownership per se, that drives the price increases.

We estimate that consumers in Bahia and Amazonas incurred a total welfare loss of approximately BRL 10.1 billion between December 2021 and June 2025. About BRL 9.2 billion of this total is a transfer from consumers to the supply chain, BRL 0.6 billion was collected as state tax revenue through the ad valorem amplification of the overcharge, and BRL 0.3 billion is deadweight loss. The estimate is conservative, because it omits indirect effects of higher energy prices on other sectors and spillovers to neighboring states in the North and Northeast, so the true welfare cost is likely higher.

A note of caution is warranted in interpreting where these consumer losses ultimately accrue. Our wholesale price analysis establishes that the overcharge originates upstream, at the refinery stage, and not from retail margins. It does not, however, establish that the new private owners earned commensurately higher profits. Under state ownership, Petrobras priced with reference to regional market conditions rather than to local operating costs alone, and may have implicitly cross-subsidized higher-cost, more isolated markets – in effect operating an informal energy policy on behalf of the government. If so, part of the post-privatization price increase could reflect the withdrawal of that cross-subsidy rather than pure rent extraction. We have no direct evidence on this question, and resolving it would require cost and profit data we do not observe. What is beyond doubt is that consumers in Bahia and Amazonas are worse off after privatization, as the prices they pay rose and the increase originates upstream.

What our results make clear is that these outcomes stem from the absence of an appropriate regulatory framework at the time of sale. This point deserves emphasis precisely

because the divestment was widely endorsed. Academics, policymakers, and the specialized press were nearly unanimous in support: Brazil's pre-privatization refining sector was an unusually concentrated monopoly with few parallels worldwide, and breaking it up was seen as an overdue structural correction. The problem was not the decision to privatize but the manner of doing so. The plants were sold as though they were competitive entities, when in fact each had been built as the single supplier of a regional market – a structure deliberately optimized for production scale economies, not for competition. Selling such assets without price regulation left the new private owners free to set prices above competitive levels.

Two policy implications follow. First, the regulatory framework must match the market structure being privatized. Where a divested asset retains local monopoly power, structural remedies alone – transferring ownership to create competition – can backfire absent accompanying safeguards such as price caps, rate-of-return regulation, or mandatory capacity-sharing arrangements. This lesson extends beyond Brazil to any divestment of geographically segmented infrastructure in markets where competition cannot be presumed to emerge from ownership change alone.

Second, several questions remain open for future research. We do not examine all products refined by the Bahia and Amazonas refineries (naphtha, aviation kerosene, and other by-products), we do not quantify the broader macroeconomic spillovers of higher energy prices, and – most importantly, given the caveat above – we do not trace where the consumer welfare loss ultimately landed: whether it accrued as profit to the new owners, as higher costs reflecting the end of cross-subsidization, or as wage gains for refinery workers. Distinguishing these channels is the natural next step in assessing the full welfare consequences of refinery privatization.

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Appendix

Parallel Trends Tests

We test the conditional parallel trends assumption using [Egami & Yamauchi \(2023\)](#), applied to the net-of-tax outcome used in our preferred specification. The following figures plot confidence intervals for pre-treatment periods (left panel) and compare treated vs. control net-of-tax prices (right panel).

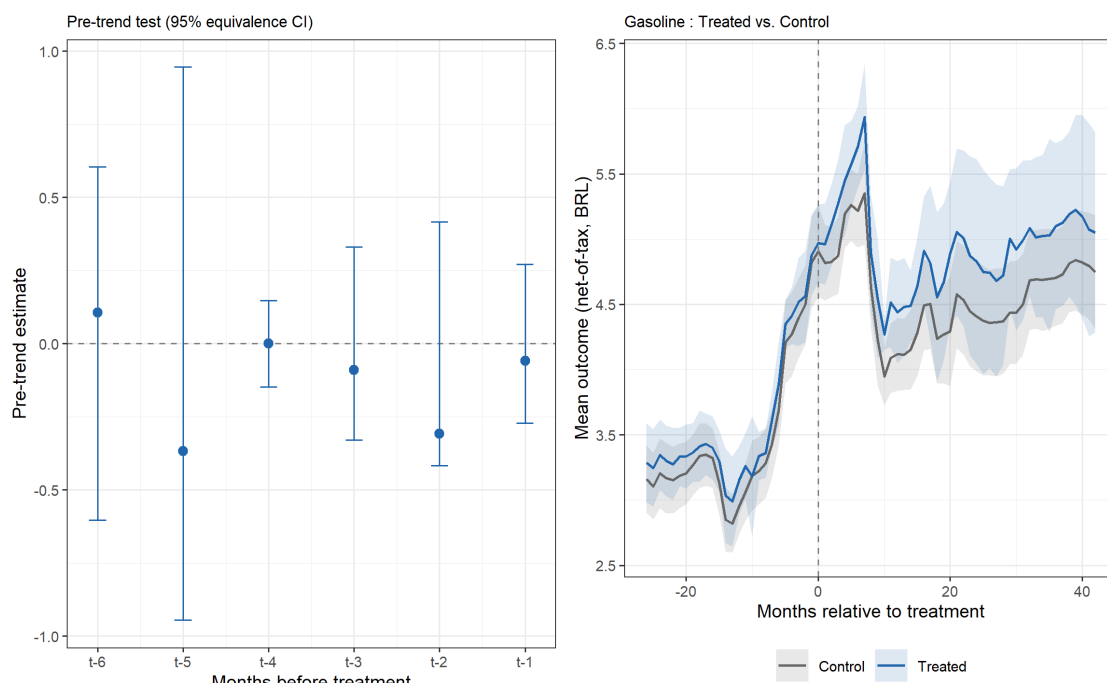


Figure 2: Gasoline: Pre-Trends and Pre/Post-Privatization Outcomes

Notes: [Egami & Yamauchi \(2023\)](#) pre-trend test on the net-of-tax outcome. Left panel: confidence intervals for pre-trends up to six months before privatization. Right panel: mean net-of-tax price (nominal BRL) for treated (Bahia and Amazonas) vs. control (all other states, excl. North and Northeast).

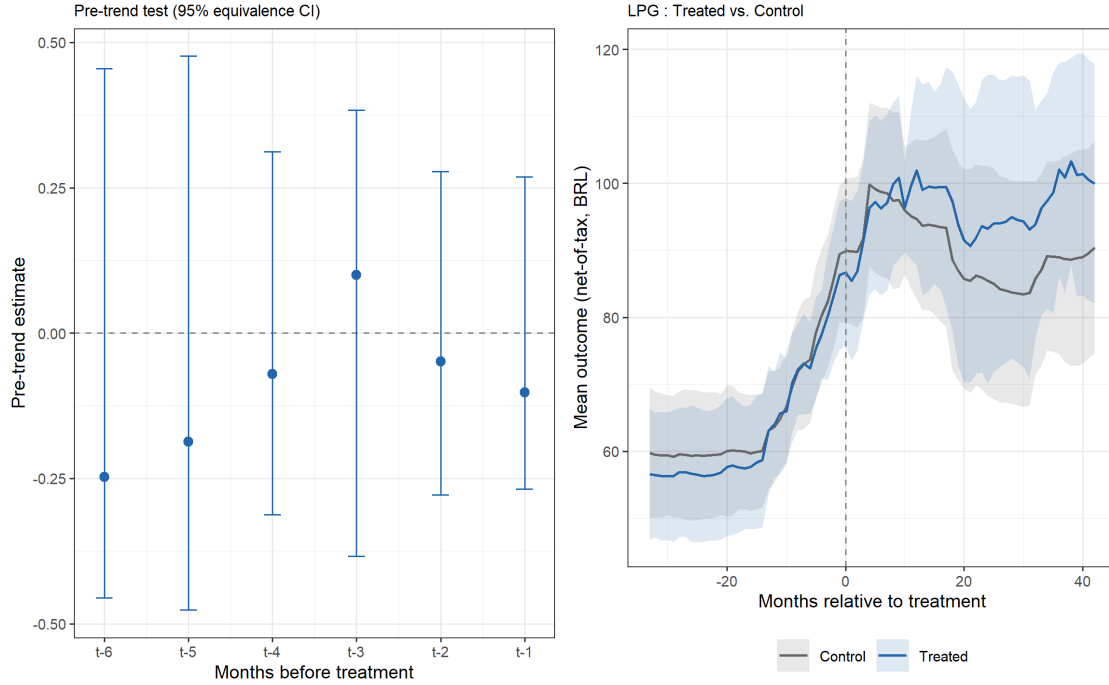


Figure 3: LPG: Pre-Trends and Pre/Post-Privatization Outcomes

Notes: [Egami & Yamauchi \(2023\)](#) pre-trend test on the net-of-tax outcome for the LPG market. See Figure 2 notes.

Fuel Taxation: Legislative Chronology and the Net-of-Tax Outcome

Fuel taxation changed repeatedly during the sample period. Table 7 documents each legal act, what it changed, and how it enters the construction of the net-of-tax outcome used in our preferred specification; Table 8 reports the state-level ad valorem ICMS rates in force before the June 2022 federal cap. Net-of-tax prices are computed as $P^{net} = P \times (1 - \tau_{st})$ in the ad valorem regimes (ICMS is levied on the tax-inclusive price) and $P^{net} = P - (\text{ad rem rate})$ in the single-stage regime, and deflated exactly as the retail prices. Two simplifications are worth noting. First, in the ad valorem regimes the statutory base was the biweekly reference price (PMPF) rather than the retail price, and we apply τ_{st} to the retail price, a transparent approximation that slightly overstates the tax when market prices run above the reference price. Second, federal PIS/Cofins changes are uniform across states and are absorbed by the week fixed effects, so no adjustment is made for them.

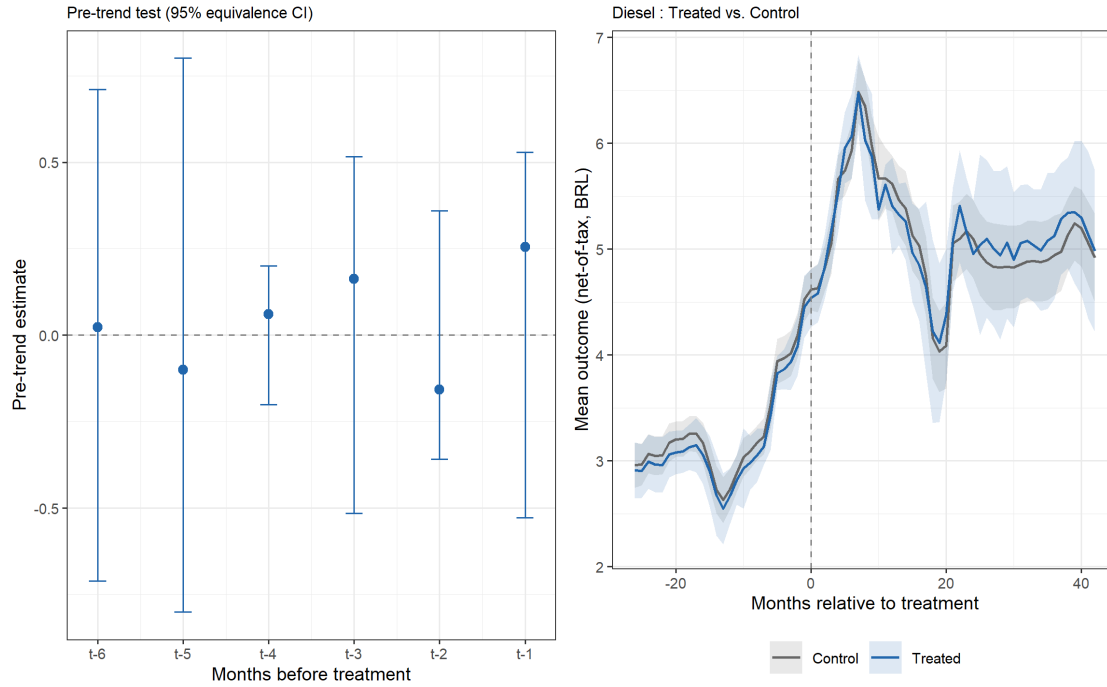


Figure 4: Diesel: Pre-Trends and Pre/Post-Privatization Outcomes

Notes: Egami & Yamauchi (2023) pre-trend test on the net-of-tax outcome for the diesel market. See Figure 2 notes.

Table 8: State ICMS Ad Valorem Rates Before the June 2022 Cap (%)

State	Gasoline	Diesel	LPG	State	Gasoline	Diesel	LPG
AM (treated)	25	18	18	PR	29	12	18
BA (treated)	28	18	12	RJ	34	12	12 ^a
DF	27	14	12	RS	25	12	12
ES	27	12	17	SC	25	12	17
GO	30	16	12	SP	25	13.3 ^b	12
MG	31	15 ^c	18	MS	30	12	17
MT	23	16	17 ^d				

Notes: Rates in force in 2021 (before Compl. Law 194/2022) for the 13 states in the estimation sample. Gasoline: project compilation (state legislation). Diesel: DIEESE (2022) (source: Fecombustíveis), cross-checked against state legislation. LPG: state legislation and official state notes. ^aExempt from Nov. 2021 (RJ Law 9,445/21). ^b12% plus the temporary 1.3 p.p. complement (Jan. 2021–Jan. 2023). ^c14% from Nov. 2021 to Mar. 2022. ^d12% from Jan. 2022 (MT Compl. Law 708/2021). With the June 2022 cap, gasoline rates fell to each state's 17–18% standard rate, diesel changed only in GO (16 → 14), and LPG rates were unaffected.

Additional Figures and Tables

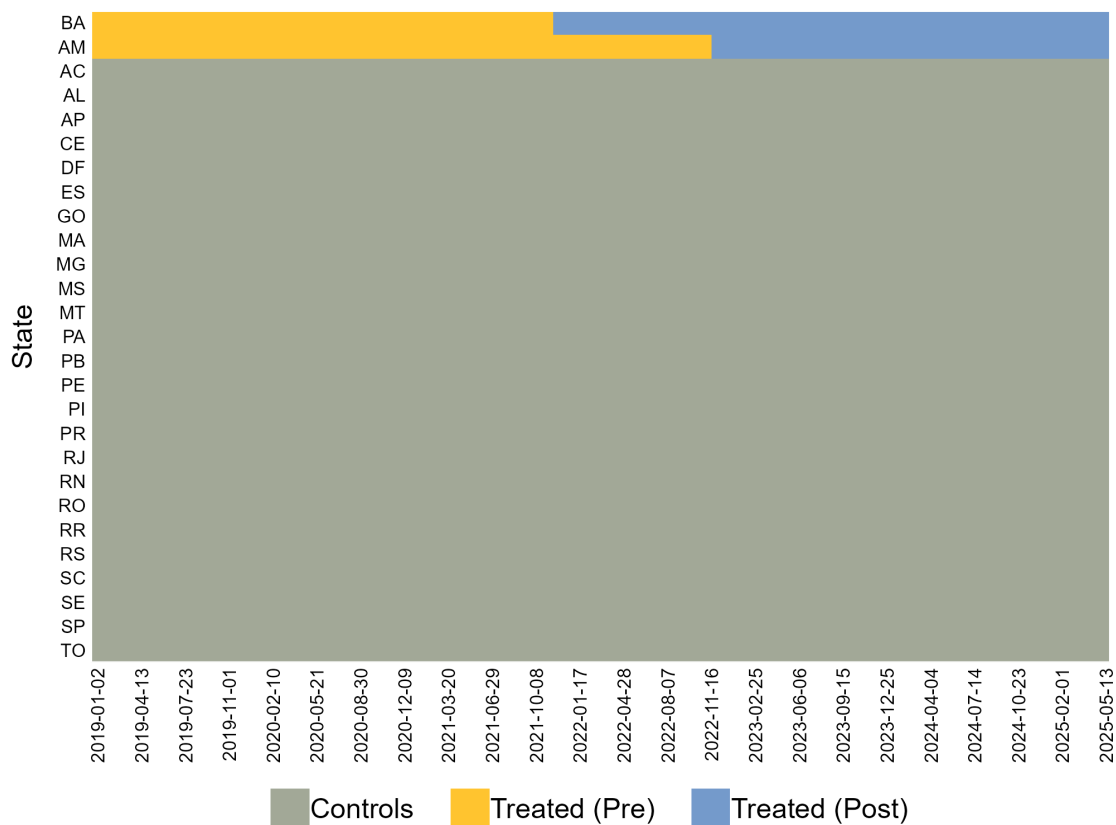


Figure 5: Treatment Status by State

Notes: Yellow cells indicate treated periods; blue cells indicate the control group; gray cells indicate states excluded from the preferred control group (North and Northeast regions). *Source:* Authors based on CADE (CADE, 2022a).

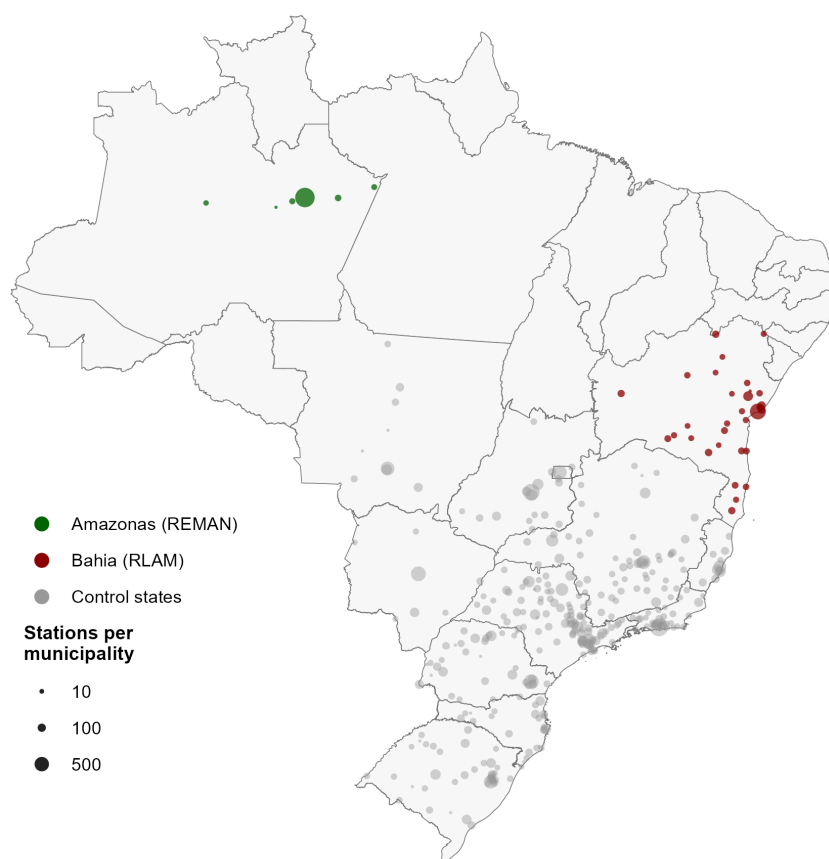


Figure 6: Geographic Distribution of the Estimation Sample

Notes: Each dot is a municipality covered by the ANP weekly price survey in our estimation sample, and dot size is proportional to the number of sampled retailers in that municipality (31,637 retailers in 366 municipalities in total). Dark green: Amazonas (1,200 retailers in 6 municipalities); dark red: Bahia (2,177 retailers in 31 municipalities); gray: control states (28,260 retailers in 329 municipalities). North and Northeast states other than the treated ones are excluded from the estimation sample. *Source:* ANP (2025).

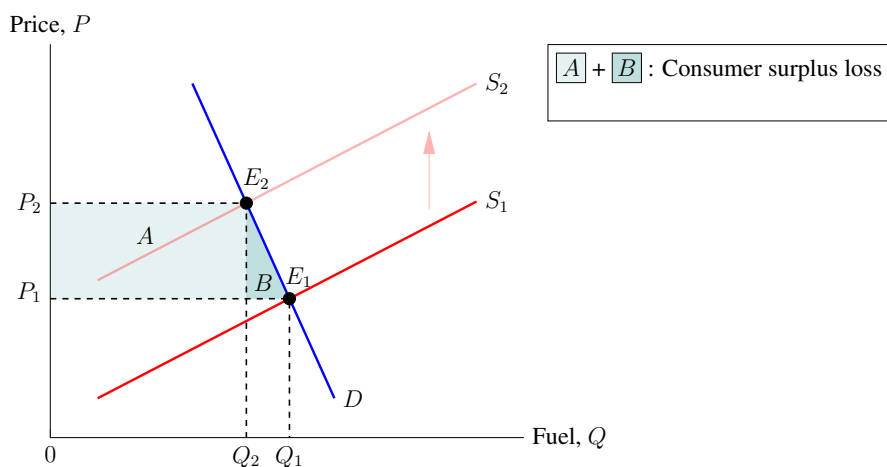


Figure 14: Consumer Surplus Loss from Privatization-Induced Price Increase

Notes: S denotes the supply curve; D the demand curve. Subscripts 1 and 2 indicate pre- and post-privatization equilibria. Area $A = \Delta P \cdot Q_2$ is the direct welfare loss; Area $B = \Delta P \cdot \Delta Q / 2$ is the indirect loss from reduced consumption. Figure adapted from Bandekar (2021).

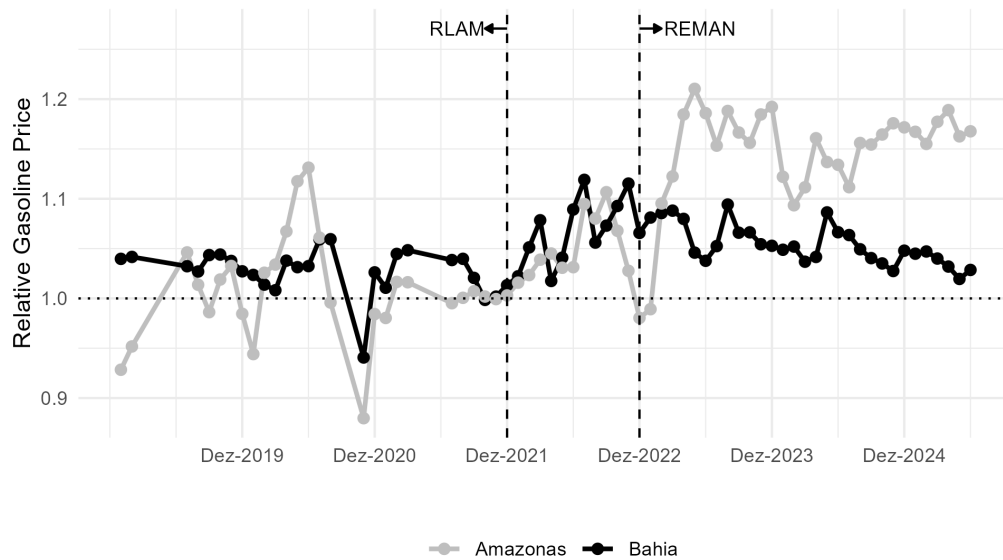


Figure 7: Relative Gasoline Price in Bahia and Amazonas, Pre and Post-Privatization
Notes: Relative prices computed as the ratio of average monthly gasoline prices in Bahia (black line) or Amazonas (gray line) to the control group average. Dashed vertical lines: the Bahia and Amazonas privatization dates. *Source:* ANP (2025).

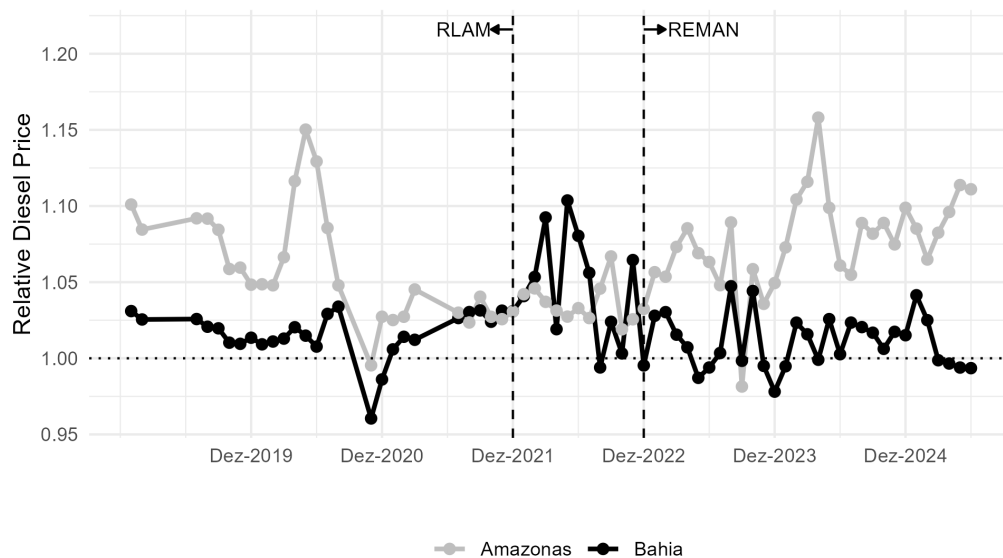
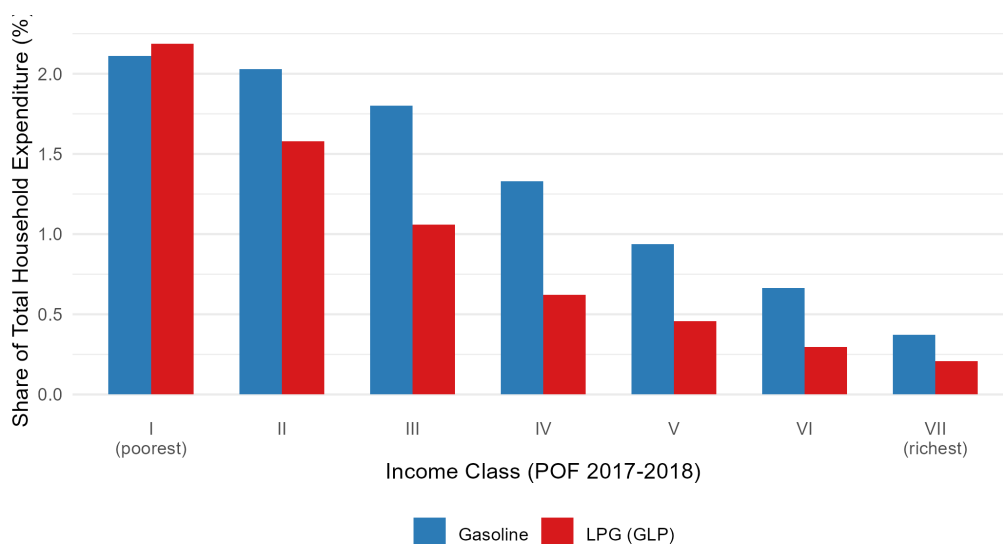


Figure 8: Relative Diesel Price in Bahia and Amazonas, Pre and Post-Privatization
Notes: See Figure 7 notes.

Table 7: Fuel-Tax Reforms During the Sample Period and Their Treatment in the Analysis

Effective date	Legal act	What changed	Treatment in the analysis
Jan. 15, 2021	SP Decree 65,253/2020	Temporary +1.3 p.p. complement on São Paulo's 12% rates: diesel 12% → 13.3% (until Jan. 14, 2023)	Diesel $\tau_{SP} = 13.3\%$ in that window
Nov. 1, 2021	MG RICMS/2002, art. 42, §36	Diesel temporarily 15% → 14% in Minas Gerais (until Mar. 31, 2022)	Diesel $\tau_{MG} = 14\%$ in that window
Nov. 3, 2021	RJ Law 9,445/2021	Rio de Janeiro exempted the 13-kg LPG cylinder from ICMS (rate was 12%)	LPG $\tau_{RJ} = 0$ from Nov. 2021
Jan. 1, 2022	MT Compl. Law 708/2021	Mato Grosso cut the LPG rate from 17% to 12%	LPG $\tau_{MT} = 12\%$ from Jan. 2022
Mar. 11, 2022	Federal Compl. Law 192/2022	Created the single-stage (<i>monofásico</i>) ICMS regime with a uniform ad rem rate per fuel (implemented in 2023); zeroed federal PIS/Cofins on diesel	Ad rem regime below; PIS/Cofins is federal and uniform → absorbed by week FE
June 23, 2022	Federal Compl. Law 194/2022	Capped state fuel ICMS at the 17–18% standard rate; binding mainly for gasoline (25–34%); among sample states it bound diesel only in GO (16% → 14%) and no LPG rate (all $\leq 18\%$)	Capped rates from June 2022 (gasoline per state in Table 8)
May 1, 2023	Conv. ICMS 199/2022	Single national ad rem rate: diesel BRL 0.9456/liter; LPG BRL 1.2571/kg	$p^{net} = P - \text{ad rem (LPG: } \times 13)$
June 1, 2023	Conv. ICMS 15/2023	Single national ad rem rate: gasoline BRL 1.22/liter	idem
Feb. 1, 2024	Conv. ICMS 172 and 173/2023	Ad rem adjusted: gasoline 1.3721; diesel 1.0635; LPG 1.4139/kg	idem
Feb. 1, 2025	Confaz 2024 acts	Ad rem adjusted: gasoline 1.47; diesel 1.12; LPG 1.39/kg	idem

Notes: Chronology compiled from [EPE \(2025\)](#), [CNT \(2024\)](#), [DIEESE \(2022\)](#), and the state legislation cited in the second column. Federal PIS/Cofins changes (zeroed in 2022, partially and then fully restored during 2023–2024) apply uniformly across states and are therefore absorbed by the week fixed effects in all specifications.



Source: IBGE, Pesquisa de Orçamentos Familiares 2017-2018 (SIDRA Table 6715).

Figure 9: Household Budget Shares of LPG and Gasoline by Income Class

Notes: Share of total household expenditure allocated to LPG and gasoline by monthly income class. The poorest households spend 10.5 times more of their budget on LPG (and 5.7 times more on gasoline) than the wealthiest. *Source:* Authors, based on the 2017–2018 Household Budget Survey (POF/IBGE, SIDRA Table 6715).

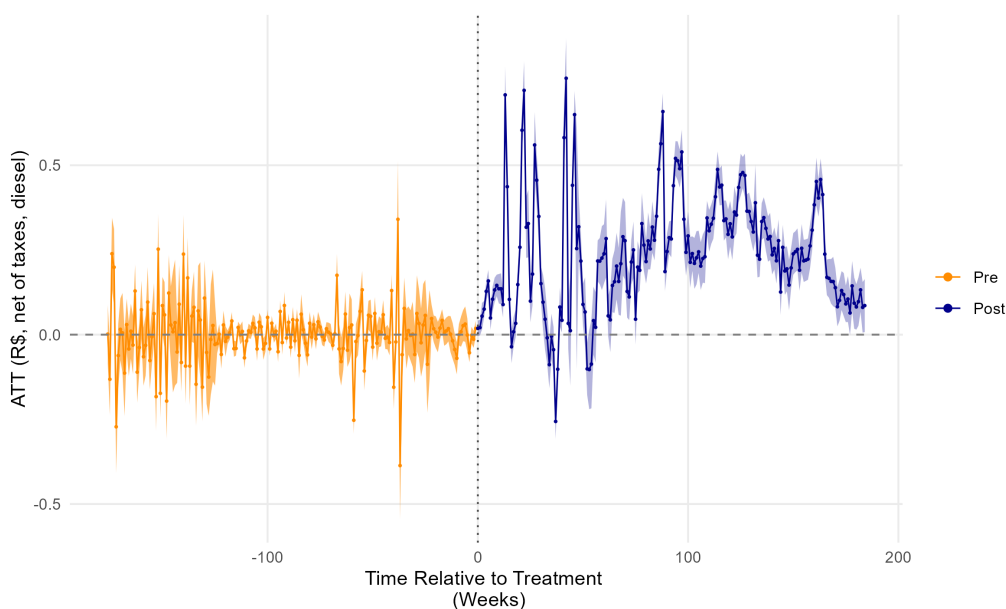


Figure 10: Diesel Overpricing Across Time (Net-of-Tax Prices)

Notes: Dynamic ATT for diesel on net-of-tax prices (preferred specification). Net of taxes, diesel overpricing is positive in both cohorts (Table 3); in retail prices the 2023 fuel-tax reform offsets the Bahia overcharge. See Figure 12 notes.

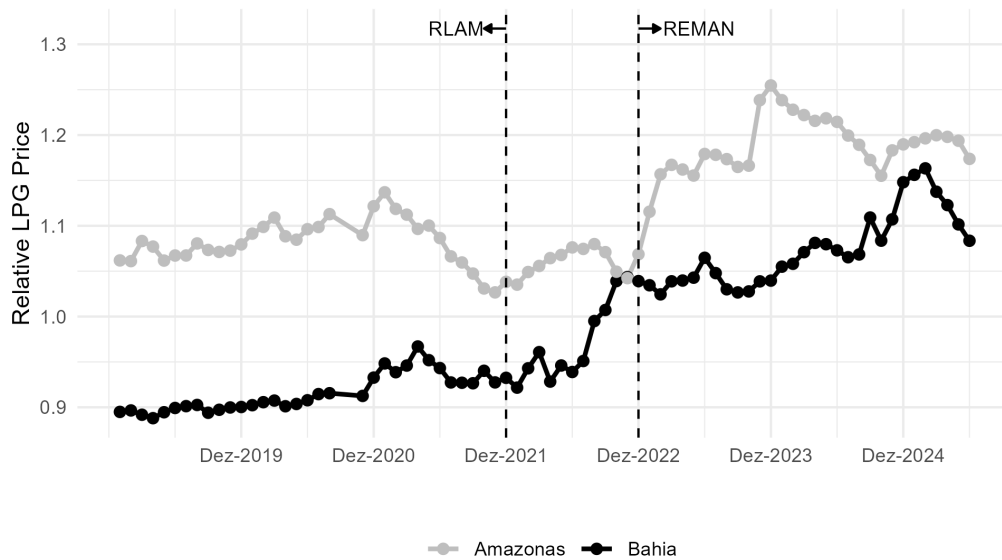


Figure 11: Relative Price of LPG in Bahia and Amazonas, Pre and Post-Privatization

Notes: The graph presents relative price comparisons, calculated by dividing the average monthly LPG price in Bahia (black line) and Amazonas (gray line) by the average price in the control group (all states excluding North and Northeast). Vertical dashed lines indicate the timing of the Bahia and Amazonas privatizations. Prices are deflated to 2025 BRL using the IPCA index. *Source:* ANP (2025).

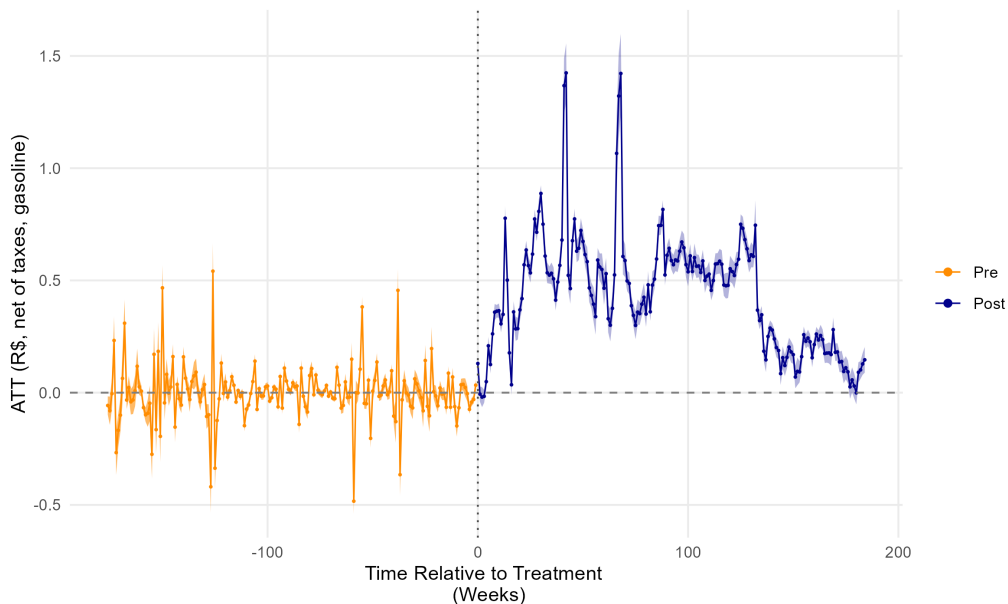


Figure 12: Gasoline Overpricing Across Time (Net-of-Tax Prices)

Notes: Dynamic ATT estimates from Callaway & Sant'Anna (2021) on net-of-tax prices (preferred specification, Table 2). Orange (pre-period) and blue (post-period) lines show point estimates with 95% confidence intervals. Horizontal dashed line at zero; vertical dotted line at treatment date. Estimates in real 2025 BRL per liter of gasoline.

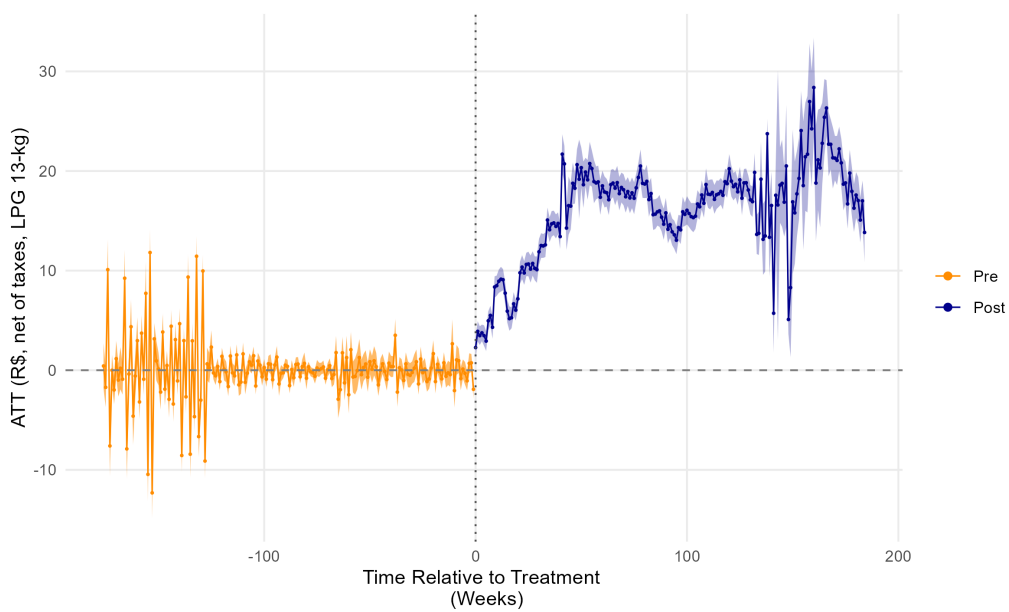


Figure 13: LPG Overpricing Across Time (Net-of-Tax Prices)

Notes: Dynamic ATT estimates from [Callaway & Sant'Anna \(2021\)](#) for LPG on net-of-tax prices (preferred specification). Estimates in real 2025 BRL per 13-kg cylinder. Orange = pre-period; blue = post-period. Horizontal dashed line at zero; vertical dotted line at treatment date.